

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	Juhi Surin	Reg. No.:	192525148
Date:	24-08-2026	Duration:	60 minutes

Code of Conduct

I Juhi Surin (192525148) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Juhi Surin

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

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Department of Computer Science and Engineering

RESEARCH PAPER REVIEW / LITERATURE SURVEY REPORT

CO5 Assessment Tool 1 – Software Engineering (CSA10)

Assigned Problem Statement (Q37): "AI-Based Digital Waste Recycling Marketplace and Resource Recovery System"

Submitted by:	Juhi Surin
Registration Number:	192525148
Course:	CSA1013 – Software Engineering
Faculty In-Charge:	Dr. Latha R
Department:	B.Tech AIML
Date of Submission:	24-08-2026

1. Introduction and Industry Context

Rapid urbanization and industrial growth have sharply increased municipal and industrial waste generation, placing pressure on governments and environmental agencies to promote circular-economy practices. Artificial Intelligence (AI), the Internet of Things (IoT), cloud computing, and digital marketplace technologies together enable more efficient waste collection, segregation, trading, and recycling, while reducing environmental impact and landfill dependency.

Existing municipal and industrial waste-management systems, however, generally lack coordination between waste generators, recycling facilities, and buyers of recyclable materials. This disconnect results in poor recycling rates, inefficient collection, and continued landfill dependence. The assigned problem statement therefore calls for an AI-powered digital marketplace that connects households, industries, recyclers, and waste-collection agencies; automatically classifies recyclable materials using AI; optimizes collection schedules; recommends nearby recycling centres; facilitates trading of recyclable materials; and generates sustainability analytics.

This report presents a literature survey of six recent, peer-reviewed research works (2023–2024) covering the three technological pillars relevant to the problem statement — AI-based waste classification, IoT-enabled smart collection and monitoring, and blockchain/digital-platform strategies for recyclable-material trading — followed by a comparative analysis, identified research gaps, and proposed future directions.

2. Review of Selected Research Papers

1. Internet of Things-Assisted Vehicle Route Optimization for Municipal Solid Waste Collection

B. Ali, M. A. Javed, A. A. K. Alharbi, S. Alotaibi, and M. Alkhathami, "Internet of Things-Assisted Vehicle Route Optimization for Municipal Solid Waste Collection," Applied Sciences, vol. 14, no. 1, p. 287, 2024, doi: 10.3390/app14010287.

Objective: To reduce municipal solid-waste collection distance and cost by selecting optimal collection routes using real-time IoT sensor data.

Methodology: IoT sensors report waste attributes (fill level, volume, toxicity) which are ranked using the TOPSIS multi-criteria decision-analysis technique to select the collection route closest to the ideal solution.

Dataset / Tools: IoT sensor network prototype; simulated municipal collection scenario evaluated in the Applied Sciences study.

Key Findings: The TOPSIS-based routing approach reduced total travel distance while simultaneously balancing several waste-related criteria, outperforming single-criterion routing methods.

2. Intelligent and Sustainable Waste Classification Model Based on Multi-Objective Beluga Whale Optimization and Deep Learning

G. I. Sayed, M. A. Elfattah, A. Darwish, and A. E. Hassanien, "Intelligent and sustainable waste classification model based on multi-objective Beluga Whale optimization and deep learning," Environmental Science and Pollution Research, vol. 31, no. 21, pp. 31492–31510, 2024.

Objective: To improve automatic waste-image classification accuracy by optimizing CNN hyperparameters for better handling of complex and composite waste materials.

Methodology: An InceptionV3 deep CNN is combined with a novel multi-objective Beluga Whale Optimization (BWO) algorithm for hyperparameter tuning, benchmarked against standard CNN architectures.

Dataset / Tools: Public waste-image benchmark datasets; InceptionV3 CNN architecture.

Key Findings: The optimized model achieved higher classification accuracy than baseline CNNs, though it still struggled with complex, composite-material waste items.

3. A Reliable and Robust Deep Learning Model for Effective Recyclable Waste Classification

M. M. Hossen et al., "A reliable and robust deep learning model for effective recyclable waste classification," IEEE Access, vol. 12, pp. 13809–13821, 2024.

Objective: To build a recyclable-waste classifier that remains accurate and robust under varied real-world imaging conditions (lighting, background clutter, occlusion).

Methodology: A deep CNN pipeline trained with extensive data augmentation and evaluated for robustness across varied image conditions, benchmarked against existing classification models.

Dataset / Tools: Recyclable-waste image dataset; deep CNN implemented and evaluated on the IEEE Access benchmark.

Key Findings: The proposed model achieved high accuracy and demonstrated stronger robustness to real-world imaging noise than prior recyclable-waste classifiers.

4. Waste Management 2.0: Leveraging Internet of Things for an Efficient and Eco-Friendly Smart City Solution

A. Bashir et al., "Waste management 2.0 leveraging internet of things for an efficient and eco-friendly smart city solution," PLOS ONE, 2024.

Objective: To design a city-wide IoT architecture for real-time waste-bin monitoring and more sustainable collection and processing.

Methodology: Ultrasonic sensors measure bin fill-level; a LoRaWAN and cellular network architecture provides connectivity; cloud analytics process the collected data for decision-making.

Dataset / Tools: Ultrasonic sensor prototype; LoRaWAN/cellular network; cloud analytics dashboard.

Key Findings: The architecture enabled reliable, city-wide, real-time bin monitoring, supporting more efficient and eco-friendly collection scheduling in a smart-city context.

5. Classification of Recyclable Waste Using Deep Learning: A Comparison of YOLO Models

S. Riyadi, A. D. Andriyani, and A. M. Masyhur, "Classification of recyclable waste using deep learning: a comparison of YOLO models," *Revue d'Intelligence Artificielle*, vol. 38, no. 4, pp. 1089–1096, 2024, doi: 10.18280/ria.380404.

Objective: To compare successive YOLO object-detection architectures for real-time recyclable-waste detection and classification.

Methodology: YOLOv5, YOLOv7 and YOLOv8 models are trained and benchmarked on the same recyclable-waste image dataset to compare detection accuracy and inference speed.

Dataset / Tools: Recyclable-waste image dataset; YOLO family of object-detection models.

Key Findings: Newer YOLO versions offered a better accuracy–speed trade-off, but overall performance remained limited by the size and diversity of available training data.

6. Reseller or Market? E-Platform Strategy Under Blockchain-Enabled Recycling

Y. Liu, D. Ma, and J. Hu, "Reseller or Market? E-platform strategy under blockchain-enabled recycling," *Expert Systems with Applications*, vol. 243, p. 122811, 2024, doi: 10.1016/j.eswa.2023.122811.

Objective: To determine which digital platform strategy (reseller model vs. open marketplace model) best incentivizes recycler participation and value distribution when recycling is blockchain-enabled.

Methodology: An analytical, game-theoretic economic model compares reseller and open-marketplace platform designs under blockchain-verified recycling transactions, validated with simulated market data.

Dataset / Tools: Analytical economic model; simulated marketplace transaction data.

Key Findings: Platform design significantly affects recycler participation and profit sharing; open-marketplace strategies were shown to improve transparency and trust among participants.

3. Comparative Literature Survey Table

The table below compares the six reviewed works on research problem, methodology, dataset/tools, key findings, limitations, and the specific research gap each leaves relative to the assigned problem statement.

Author / Year	Research Problem / Focus	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Ali et al. (2024)	IoT-based municipal solid waste (MSW) collection route selection	IoT sensors + TOPSIS multi-criteria decision analysis for route ranking (toxicity, volume, time)	IoT sensor network; MDPI Applied Sciences case simulation	Reduced route distance while balancing multiple waste attributes simultaneously	Tested only in simulation; no live fleet deployment or AI-based prediction of fill levels	No integration of predictive AI classification with routing decisions
Sayed et al. (2024)	Automated waste image classification	InceptionV3 CNN tuned with multi-objective Beluga Whale Optimization (BWO)	Public waste-image datasets (TrashNet-type); benchmarked against VGG/ResNet	Improved classification accuracy and reduced misclassification of mixed-material waste	High computational cost of optimization; struggles with composite/multi-material items	Lack of lightweight models deployable on low-cost edge/marketplace devices
Hossen et al. (2024)	Reliable recyclable-waste classification for real-world conditions	Deep CNN pipeline with data augmentation and robustness testing	Recyclable waste image dataset; IEEE Access benchmark	High accuracy and robustness under varied lighting/background conditions	Focuses only on classification; no linkage to collection or marketplace logistics	No end-to-end system connecting classification output to trading/recovery workflow
Bashir et al. (2024)	Smart-city IoT architecture for waste monitoring	Ultrasonic fill-level sensors + LoRaWAN/cellular connectivity + cloud analytics	Prototype smart-bin sensors; cloud dashboard	Enabled real-time bin monitoring and more eco-friendly collection scheduling	Limited to monitoring/alerts; lacks AI-driven material classification or stakeholder trading	No mechanism for connecting recyclers, generators and buyers digitally
Riyadi et al. (2024)	Comparative evaluation of YOLO models for recyclable-	Benchmarking YOLOv5/v7/v8 object-detection architectures	Recyclable waste image dataset (Revue d'Intelligence Artificielle)	Identified accuracy-speed trade-offs across YOLO versions for real-time sorting	Evaluated only detection accuracy; no field deployment or marketplace integration	Absence of large, standardized, regionally diverse waste datasets for training

Auth or / Year	Research Problem / Focus	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
	e waste detection					
Liu, Ma & Hu (2024)	Platform-strategy design for blockchain-enabled recycling e-commerce	Analytical/game-theoretic modelling of reseller vs. open-market platform strategies	Economic model validated with simulation data (Expert Systems with Applications)	Showed how platform design choice affects recycler participation and profit sharing	Purely economic/strategic model; no AI classification or IoT data integration	No unified framework combining AI classification, IoT collection and blockchain trading

4. Critical Analysis and Identified Research Gap

Across the reviewed literature, three technology streams have matured largely in isolation. AI/deep-learning classification studies (Sayed et al., 2024; Hossen et al., 2024; Riyadi et al., 2024) report high accuracy on benchmark datasets but focus purely on image-level classification, without linking predictions to downstream collection or trading actions. IoT-based collection and monitoring studies (Ali et al., 2024; Bashir et al., 2024) improve routing and bin-monitoring efficiency but treat waste only as sensor-detected volume or toxicity, with no material-level AI classification or connection to a marketplace. The blockchain/platform-strategy study (Liu, Ma and Hu, 2024) addresses fair value-sharing and transparency for recyclable trading but does not incorporate AI classification or IoT-based collection data into its model.

The consolidated research gap is therefore the absence of a single, unified system that integrates: (i) AI-based automatic classification of recyclable materials, (ii) IoT-driven real-time collection scheduling and bin monitoring, and (iii) a transparent digital marketplace mechanism for connecting waste generators, recyclers, and buyers. Existing works also share common limitations — reliance on small or non-diverse datasets, difficulty classifying composite/mixed-material waste, evaluation limited to simulation rather than field deployment, and a lack of sustainability-analytics reporting for stakeholders — all of which directly correspond to the objectives of the assigned problem statement.

5. Future Research Directions and Relevance to the Problem Statement

- Develop a unified architecture that fuses AI-based waste classification, IoT sensor data, and a blockchain-backed marketplace layer into a single platform, directly addressing Objectives 1–3 of the assigned problem.

- Build larger, regionally diverse, multi-material waste-image datasets to improve classification of composite and mixed recyclables, addressing the dataset limitation common to Sayed et al. (2024), Hossen et al. (2024), and Riyadi et al. (2024).
- Extend IoT routing models (Ali et al., 2024; Bashir et al., 2024) with AI-predicted material value and quantity, enabling smarter, recycler-aware collection scheduling rather than volume-only routing.
- Adopt transparent, blockchain-verified transaction mechanisms informed by Liu, Ma and Hu (2024) to build trust among households, industries, recyclers, and collection agencies within the marketplace.
- Incorporate sustainability-analytics dashboards that translate classification, collection, and trading data into landfill-reduction and circular-economy performance indicators for stakeholders and regulators.
- Pilot the integrated system in real municipal or campus environments (moving beyond simulation) to validate collection-efficiency, classification-accuracy, and marketplace-adoption outcomes together.

6. Conclusion

The literature confirms strong, independently mature progress in AI-based waste classification, IoT-enabled smart collection, and blockchain-enabled recycling platforms, but no reviewed work combines all three into one coordinated system. This validates the need for the assigned AI-Based Digital Waste Recycling Marketplace and Resource Recovery System, which can build on the classification accuracy, routing efficiency, and platform-trust mechanisms demonstrated in the reviewed papers while closing the integration gap identified above.

7. References

- [1] B. Ali, M. A. Javed, A. A. K. Alharbi, S. Alotaibi, and M. Alkhathami, "Internet of Things-Assisted Vehicle Route Optimization for Municipal Solid Waste Collection," *Applied Sciences*, vol. 14, no. 1, p. 287, 2024, doi: 10.3390/app14010287.
- [2] G. I. Sayed, M. A. Elfattah, A. Darwish, and A. E. Hassanien, "Intelligent and sustainable waste classification model based on multi-objective Beluga Whale optimization and deep learning," *Environmental Science and Pollution Research*, vol. 31, no. 21, pp. 31492–31510, 2024.
- [3] M. M. Hossen et al., "A reliable and robust deep learning model for effective recyclable waste classification," *IEEE Access*, vol. 12, pp. 13809–13821, 2024.
- [4] A. Bashir et al., "Waste management 2.0 leveraging internet of things for an efficient and eco-friendly smart city solution," *PLOS ONE*, 2024.
- [5] S. Riyadi, A. D. Andriyani, and A. M. Masyhur, "Classification of recyclable waste using deep learning: a comparison of YOLO models," *Revue d'Intelligence Artificielle*, vol. 38, no. 4, pp. 1089–1096, 2024, doi: 10.18280/ria.380404.

[6] Y. Liu, D. Ma, and J. Hu, "Reseller or Market? E-platform strategy under blockchain-enabled recycling," *Expert Systems with Applications*, vol. 243, p. 122811, 2024, doi: 10.1016/j.eswa.2023.122811.